

Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

CHAPTER – II

IDEATION PHASE

1.2 – Purpose of the Project

With the rising demand for power, urbanization, industrial development, and changes in the consumption pattern, there is an immense need for efficient management of the same. To ensure sustainable development and optimal management of resources, it is crucial to have knowledge of how much power was consumed in different states, regions, cities, and time periods. With the rise in the number of records being collected every year about electricity consumption, it becomes quite challenging to identify trends, seasonal changes, regional differences, or any abnormality in consumption patterns.

There are many organizations and individuals who face difficulties while analyzing data regarding electricity consumption due to the lack of a platform for conducting an insightful analysis. Not having access to tools for conducting an analysis makes it hard to predict and plan for the future regarding energy consumption and distribution. Furthermore, population, economic, climate change, and even sudden events, like a lockdown caused by the coronavirus pandemic, greatly affect electricity consumption. Therefore, it is necessary to conduct an analysis of electricity consumption patterns before and after such events.

Consequently, the need for a data-driven solution capable of collecting, processing, visualizing, and analyzing electricity consumption data arises. The current project, titled "Plugging into the Future: An Exploration of Electricity Consumption Patterns," uses Tableau to develop dashboards and stories that will give valuable insight into yearly, monthly, quarterly, regional, and state-wise electricity consumption patterns. The main objective of this project includes the identification of trends in the consumption of electricity; comparison of consumption patterns for various regions; and highlighting the best or worst electricity consumers.